

The biconstrained function ψ of Chudnovsky, Hompe, Scott, Seymour and Spirkł is not symmetric:

$$\psi(2/7, 5/7) \leq 23/28 < 6/7 = \psi(5/7, 2/7)$$

github.com/graph-theory-AI

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The constructions and proofs were found by the language model `gpt-5.6-sol` in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 1902.10878__01; original writeup in `attacks/1902.10878__01/output.md`). (2) An independent adversarial referee agent (`claude-fable-5`) checked the definitions against the arXiv v3 TeX source of the paper, re-derived every step of the rigidity argument, rebuilt the 42-vertex certificate independently from the prose and verified it exactly (together with the $q = 9, 11$ members of the family and the cyclic constructions for $q = 3, \dots, 13$), and machine-checked the parity obstruction by exhaustive enumeration of all matchings of K_q for $q \leq 13$; its verdict was **CONFIRMED**. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by `claude-fable-5.1`: the text was reorganised with full proofs in [Appendix A](#), the $q = 7$ certificate was written out as explicit tables ([Table 1](#)), and [Section 4](#) adds context taken from the referee's report (consistency with the theorems of the source paper, the role of $q \geq 7$, scope). No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

For $x, y \in (0, 1]$, Chudnovsky, Hompe, Scott, Seymour and Spirkł define $\psi(x, y)$ as the largest z such that in every graph with an (x, y) -biconstrained tripartition (A, B, C) some vertex of C is joined by two-edge paths to at least $z|A|$ vertices of A . They prove that the one-sided analogue ϕ is symmetric, $\phi(x, y) = \phi(y, x)$, and ask whether the same holds for ψ . We show that it does not: for every odd $q \geq 7$,

$$\psi\left(\frac{2}{q}, \frac{q-2}{q}\right) \leq 1 - \frac{2(q-2)}{q(q+1)} < 1 - \frac{1}{q} = \psi\left(\frac{q-2}{q}, \frac{2}{q}\right).$$

The value $\psi((q-2)/q, 2/q) = 1 - 1/q$ is determined exactly by a rigidity argument ending in a parity obstruction; the upper bound on $\psi(2/q, (q-2)/q)$ comes from an explicit triple with $|A| = q(q+1)/2$ and $|B| = |C| = q$. For $q = 7$ this is a 42-vertex graph with $\psi(2/7, 5/7) \leq 23/28 < 24/28 = \psi(5/7, 2/7)$.

1 Introduction

All graphs are finite and simple. Following [1], a *tripartition* of a graph G is a partition (A, B, C) of $V(G)$ into three nonempty stable sets such that there are no edges between A and C . For $x, y \in (0, 1]$ the tripartition is (x, y) -*constrained* if every vertex of A has at least $x|B|$ neighbours in B and every vertex of B has at least $y|C|$ neighbours in C , and it is (x, y) -*biconstrained* if in addition every vertex

of B has at least $x|A|$ neighbours in A and every vertex of C has at least $y|B|$ neighbours in B . In symbols, writing $N_X(v)$ for the set of neighbours of v in X , the four biconstrained conditions read

$$\begin{aligned} |N_B(a)| &\geq x|B| \quad (a \in A), & |N_A(b)| &\geq x|A| \quad (b \in B), \\ |N_C(b)| &\geq y|C| \quad (b \in B), & |N_B(c)| &\geq y|B| \quad (c \in C). \end{aligned} \quad (1)$$

For $c \in C$ let $N_A^2(c)$ be the set of vertices of A at distance exactly two from c ; since the parts are stable and there are no A – C edges, this is

$$N_A^2(c) = R(c) := \{a \in A : N_B(a) \cap N_B(c) \neq \emptyset\}.$$

The paper defines $\phi(x, y)$ as the maximum z such that every (x, y) -constrained tripartition has a vertex $c \in C$ with $|N_A^2(c)| \geq z|A|$, and $\psi(x, y)$ in the same way for (x, y) -biconstrained tripartitions (the maximum exists, [1, Theorem 2.1]); equivalently

$$\psi(x, y) = \inf_{(G, A, B, C)} \max_{c \in C} \frac{|R(c)|}{|A|}, \quad (2)$$

the infimum over all (x, y) -biconstrained tripartitions. Theorem 2.3 of [1] states that $\phi(x, y) = \phi(y, x)$ for all $x, y \in (0, 1]$, proved by a linear-programming argument. Immediately afterwards the authors write: “we have not been able to prove an analogue of [Theorem 2.3] for the biconstrained case, or for the exact case, although we have no counterexample for either one.” The catalog records this as:

Question 1.1 ([1], remark after Theorem 2.3). Does $\psi(x, y) = \psi(y, x)$ hold for all $x, y \in (0, 1]$?

We answer this negatively.

Theorem 1.2 (Main theorem). *For every odd integer $q \geq 7$,*

$$\psi\left(\frac{2}{q}, \frac{q-2}{q}\right) \leq 1 - \frac{2(q-2)}{q(q+1)} < 1 - \frac{1}{q} = \psi\left(\frac{q-2}{q}, \frac{2}{q}\right).$$

Corollary 1.3. $\psi(2/7, 5/7) \leq \frac{23}{28} < \frac{24}{28} = \frac{6}{7} = \psi(5/7, 2/7)$. *In particular the answer to Question 1.1 is negative.*

Proof. Take $q = 7$ in Theorem 1.2: $1 - \frac{2 \cdot 5}{7 \cdot 8} = 1 - \frac{10}{56} = \frac{23}{28}$ and $1 - \frac{1}{7} = \frac{24}{28}$. $\square$

Theorem 1.2 combines two statements, proved in Sections 2 and 3: the exact value $\psi((q-2)/q, 2/q) = 1 - 1/q$ for all odd $q \geq 3$ (Lemma 2.1), and the upper bound $\psi(2/q, (q-2)/q) \leq 1 - 2(q-2)/(q(q+1))$ for odd $q \geq 7$ (Proposition 3.1). The theorem follows because

$$\frac{2(q-2)}{q(q+1)} - \frac{1}{q} = \frac{2q-4-(q+1)}{q(q+1)} = \frac{q-5}{q(q+1)} > 0 \quad (q > 5). \quad (3)$$

Idea of proof. Both values are governed by the same rigidity: if $a \in A$ is *not* reached by $c \in C$, then $N_B(a)$ and $N_B(c)$ are disjoint, and when $x + y = 1$ the degree conditions force $N_B(a) = B \setminus N_B(c)$ with both degrees tight. For $(x, y) = ((q-2)/q, 2/q)$ this rigidity propagates: if every c missed more than $|A|/q$ vertices, the B -neighbourhoods S_i of C -vertices would each carry exactly a $2/q$ fraction of C , and the B -side degree bound would force these S_i to be pairwise disjoint, so that m of them cover C exactly, giving $2m = q$, which is impossible for odd q . A cyclic construction on $3q$ vertices shows $1 - 1/q$ is attained. For $(x, y) = (2/q, (q-2)/q)$ the roles flip: now A -vertices have tiny neighbourhoods S_i of size two, and there is no parity obstruction to choosing $m = (q+3)/2$ two-element sets S_i in a q -set B that “cover” B with the right multiplicities (disjoint pairs plus a triangle). Giving each S_i a block of $q-2$ A -vertices, adding three A -vertices complete to B to lift the B -side degrees, and letting each C -vertex miss exactly one S_i produces a triple in which every C -vertex misses exactly $q-2$ of the $q(q+1)/2$ vertices of A . [full proof]

2 The exact value of $\psi((q-2)/q, 2/q)$

Lemma 2.1. For every odd integer $q \geq 3$, $\psi\left(\frac{q-2}{q}, \frac{2}{q}\right) = 1 - \frac{1}{q}$.

Idea of proof. Lower bound. Suppose every $c \in C$ reaches fewer than $\frac{q-1}{q}|A|$ vertices. For an unreached pair (a, c) , disjointness of $N_B(a), N_B(c)$ and $|N_B(a)| + |N_B(c)| \geq |B|$ force $N_B(a) = B \setminus N_B(c)$ and $|N_B(c)| = \frac{2}{q}|B|$. Counting B - C edges then forces $|N_C(b)| = \frac{2}{q}|C|$ for every b . Group C by its B -neighbourhoods $S_1, \dots, S_m$, with fractions β_i of C , and let A_i be the A -vertices with neighbourhood $B \setminus S_i$, with fractions $\alpha_i > 1/q$ of A . For $b \in B$, the indices i with $b \in S_i$ have $\sum \beta_i = 2/q$ and, because b has at most $\frac{2}{q}|A|$ non-neighbours in A , $\sum \alpha_i \leq 2/q$; hence each b lies in exactly one S_i , each $\beta_i = 2/q$, and $1 = \sum \beta_i = 2m/q$, contradicting the parity of q . *Upper bound.* On $A = B = C = \mathbb{Z}_q$ let $N_B(c_i) = \{b_i, b_{i+1}\}$ and $N_B(a_i) = B \setminus \{b_i, b_{i+1}\}$; each c_i misses exactly a_i . [\[full proof\]](#)

3 The construction for $\psi(2/q, (q-2)/q)$

Let $q \geq 7$ be odd and put

$$h = \frac{q-3}{2} (\geq 2), \quad m = h+3 = \frac{q+3}{2}.$$

Let B consist of h pairwise disjoint pairs $P_1, \dots, P_h$ together with three further vertices r_{12}, r_{13}, r_{23} , so $|B| = 2h+3 = q$. Define m distinct two-element subsets of B :

$$S_i = P_i \quad (1 \leq i \leq h), \quad S_{h+1} = \{r_{12}, r_{13}\}, \quad S_{h+2} = \{r_{12}, r_{23}\}, \quad S_{h+3} = \{r_{13}, r_{23}\}.$$

- A : for each $i \in \{1, \dots, m\}$ a set A_i of $q-2$ vertices, each with $N_B(a) = S_i$; and a set U of three vertices, each with $N_B(u) = B$. Thus $|A| = m(q-2) + 3 = q(q+1)/2$.
- C : for each $1 \leq i \leq h$ two vertices of type i , and for each $i \in \{h+1, h+2, h+3\}$ one vertex of type i ; a vertex c of type i has $N_B(c) = B \setminus S_i$. Thus $|C| = 2h+3 = q$.

The graph G_q has vertex set $A \cup B \cup C$ and exactly the A - B and B - C edges just described; so (A, B, C) is a tripartition (three nonempty stable sets, no A - C edges).

Proposition 3.1. For every odd $q \geq 7$, (A, B, C) is a $(2/q, (q-2)/q)$ -biconstrained tripartition of G_q with $|A| = q(q+1)/2$, $|B| = |C| = q$, and every $c \in C$ satisfies $|A \setminus R(c)| = q-2$. Consequently

$$\psi\left(\frac{2}{q}, \frac{q-2}{q}\right) \leq 1 - \frac{q-2}{q(q+1)/2} = 1 - \frac{2(q-2)}{q(q+1)}.$$

Idea of proof. Degrees: A_i -vertices have $2 = \frac{2}{q}|B|$ neighbours in B ; a vertex of a pair P_i is adjacent to $A_i \cup U$, i.e. to $q+1 = \frac{2}{q}|A|$ vertices; each r_{jk} lies in two of the last three S_i and has $2(q-2)+3 \geq q+1$; every c has $q-2 = \frac{q-2}{q}|B|$ neighbours in B ; every b is omitted by exactly two C -vertices, hence has $q-2 = \frac{q-2}{q}|C|$ neighbours in C . Reach: a c of type i misses A_i (whose neighbourhood is S_i) and reaches every A_j , $j \neq i$ (the distinct two-sets satisfy $S_j \not\subseteq S_i$) and U . [\[full proof\]](#)

The certificate for $q = 7$. Here $h = 2$, $m = 5$, $|A| = 28$, $|B| = |C| = 7$. Write $B = \{1, \dots, 7\}$ with $P_1 = \{1, 2\}$, $P_2 = \{3, 4\}$, $r_{12} = 5$, $r_{13} = 6$, $r_{23} = 7$, so

$$S_1 = \{1, 2\}, \quad S_2 = \{3, 4\}, \quad S_3 = \{5, 6\}, \quad S_4 = \{5, 7\}, \quad S_5 = \{6, 7\}.$$

[Table 1](#) lists every vertex of A and C with its neighbourhood in B ; there are no other edges. Every C -vertex misses exactly the five A -vertices of its own type and reaches the other 23, so $\psi(2/7, 5/7) \leq 23/28$.

part	vertices	$N_B(\cdot)$	count
A	$a_{1,1}, \dots, a_{1,5}$	$\{1, 2\}$	5
	$a_{2,1}, \dots, a_{2,5}$	$\{3, 4\}$	5
	$a_{3,1}, \dots, a_{3,5}$	$\{5, 6\}$	5
	$a_{4,1}, \dots, a_{4,5}$	$\{5, 7\}$	5
	$a_{5,1}, \dots, a_{5,5}$	$\{6, 7\}$	5
	u_1, u_2, u_3	$\{1, 2, 3, 4, 5, 6, 7\}$	3
C	c_1, c'_1	$\{3, 4, 5, 6, 7\}$	2
	c_2, c'_2	$\{1, 2, 5, 6, 7\}$	2
	c_3	$\{1, 2, 3, 4, 7\}$	1
	c_4	$\{1, 2, 3, 4, 6\}$	1
	c_5	$\{1, 2, 3, 4, 5\}$	1

Table 1: The 42-vertex $(2/7, 5/7)$ -biconstrained triple, $B = \{1, \dots, 7\}$. Degree check: each $a_{i,j}$ has $2 = \frac{2}{7} \cdot 7$ neighbours in B ; each $b \in \{1, 2, 3, 4\}$ has $5 + 3 = 8 = \frac{2}{7} \cdot 28$ neighbours in A and each $b \in \{5, 6, 7\}$ has $10 + 3 = 13$; each c has $5 = \frac{5}{7} \cdot 7$ neighbours in B ; each b is missing from exactly two rows of the C block, so has $5 = \frac{5}{7} \cdot 7$ neighbours in C . Reach check: c of type i has a common B -neighbour with every A -vertex except the five with neighbourhood S_i .

4 Remarks

Remark 4.1 (Why $q \geq 7$). By (3) the two bounds of [Theorem 1.2](#) differ by $(q - 5)/(q(q + 1))$, which vanishes at $q = 5$ and is negative at $q = 3$. At $q = 3$ both sides equal $2/3$, in agreement with the values $\psi(1/3, 2/3) = \psi(2/3, 1/3) = 2/3$ that follow from the source paper’s Theorems 2.2 and the “2 + 3” bounds (as the referee notes); the construction of [Section 3](#) needs $h \geq 2$ anyway, i.e. $q \geq 7$.

Remark 4.2 (Consistency with the source paper). The referee ([Appendix B](#)) checked the two values against every result of [1]. The trivial bounds $\max(x, y) \leq \psi(x, y) \leq (\lceil kx \rceil + \lceil ky \rceil - 1)/k$ give, with $k = 7$, $\psi \in [5/7, 6/7]$ for both $(5/7, 2/7)$ and $(2/7, 5/7)$; the value $6/7$ attains the upper bound, and $23/28$ lies strictly inside. The paper’s Theorem giving $\psi(x, y) \geq 3/4$ for $x \geq 1/4$, $y > 2/3$ yields $\psi(2/7, 5/7) \geq 3/4 = 21/28$, so $\psi(2/7, 5/7) \in [21/28, 23/28]$; its exact value is not determined here. The paper’s partial symmetry result (Theorem 10.2: $\psi(x, y) = x$ iff $\psi(y, x) = x$) is not contradicted, since neither value equals $\max(x, y)$.

Remark 4.3 (Scope). The symmetry $\phi(x, y) = \phi(y, x)$ (Theorem 2.3 of [1]) is unaffected. The construction uses vertices of A complete to B , so it says nothing about the paper’s *exact* variant (every degree condition holding with equality), whose symmetry question remains open.

Remark 4.4 (Computational checks). The referee rebuilt both constructions from the prose and verified, with exact rational arithmetic, all degree conditions and all second neighbourhoods for $q = 7, 9, 11$ (every c reaches $23/28, 38/45, 57/66$ of A) and for the cyclic construction for $q = 3, \dots, 13$; it also verified the finite core of the parity step by enumerating all matchings of K_q , $q \leq 13$. Scripts: [verification/scripts/1902.10878__01/](#).

Remark 4.5 (Novelty). The referee’s literature search (September 2026) found no resolution of [Question 1.1](#); the published version of [1] still lists it as open, and the exact value $\psi((q-2)/q, 2/q) = 1 - 1/q$ does not appear there. This has not been checked by a human.

A Full proofs

Lemma 2.1. For every odd integer $q \geq 3$, $\psi\left(\frac{q-2}{q}, \frac{2}{q}\right) = 1 - \frac{1}{q}$.

Proof. Fix odd $q \geq 3$ and write $x = (q-2)/q$, $y = 2/q$, so $x + y = 1$.

Lower bound. Let (A, B, C) be an (x, y) -biconstrained tripartition of a graph G , and suppose for a contradiction that

$$|R(c)| < \frac{q-1}{q}|A| \quad \text{for every } c \in C. \quad (4)$$

Then every $c \in C$ has more than $|A|/q > 0$ vertices of A outside $R(c)$. Let $a \notin R(c)$. Then $N_B(a) \cap N_B(c) = \emptyset$, so by (1)

$$|B| \geq |N_B(a)| + |N_B(c)| \geq \frac{q-2}{q}|B| + \frac{2}{q}|B| = |B|.$$

Equality holds throughout, hence

$$|N_B(a)| = \frac{q-2}{q}|B|, \quad |N_B(c)| = \frac{2}{q}|B|, \quad N_B(a) = B \setminus N_B(c) \quad \text{whenever } a \notin R(c). \quad (5)$$

Since every c has some unreachable a , (5) gives $|N_B(c)| = \frac{2}{q}|B|$ for every $c \in C$, so $e(B, C) = \sum_{c \in C} |N_B(c)| = \frac{2}{q}|B||C|$. On the other hand $e(B, C) = \sum_{b \in B} |N_C(b)|$ with every term at least $\frac{2}{q}|C|$ by (1); equality of the totals forces

$$|N_C(b)| = \frac{2}{q}|C| \quad \text{for every } b \in B. \quad (6)$$

Let $S_1, \dots, S_m$ be the distinct sets occurring as $N_B(c)$, $c \in C$, and put

$$C_i = \{c \in C : N_B(c) = S_i\}, \quad \beta_i = \frac{|C_i|}{|C|}, \quad A_i = \{a \in A : N_B(a) = B \setminus S_i\}, \quad \alpha_i = \frac{|A_i|}{|A|}.$$

Then $\beta_i > 0$, $\sum_i \beta_i = 1$, and the sets A_i are pairwise disjoint because the S_i are distinct. For $c \in C_i$ we claim $A \setminus R(c) = A_i$: if $a \notin R(c)$ then $N_B(a) = B \setminus S_i$ by (5), so $a \in A_i$; conversely if $a \in A_i$ then $N_B(a) \cap S_i = \emptyset$, so $a \notin R(c)$. Hence (4) gives

$$\alpha_i > \frac{1}{q} \quad (1 \leq i \leq m). \quad (7)$$

For $b \in B$ let $I(b) = \{i : b \in S_i\}$. Since $N_C(b) = \bigcup_{i \in I(b)} C_i$ (a disjoint union), (6) gives

$$\sum_{i \in I(b)} \beta_i = \frac{2}{q}, \quad (8)$$

in particular $I(b) \neq \emptyset$. If $i \in I(b)$, every $a \in A_i$ has $N_B(a) = B \setminus S_i \not\ni b$, so A_i is entirely non-adjacent to b . By (1), b has at least $\frac{q-2}{q}|A|$ neighbours in A , hence at most $\frac{2}{q}|A|$ non-neighbours, and since the A_i are disjoint,

$$\sum_{i \in I(b)} \alpha_i \leq \frac{2}{q}. \quad (9)$$

By (7), two terms of this sum would already exceed $2/q$; so $|I(b)| = 1$ for every $b \in B$, and if $I(b) = \{i\}$ then (8) gives $\beta_i = 2/q$. Every index i occurs in this way: $S_i = N_B(c)$ for some c has size $\frac{2}{q}|B| > 0$, and any $b \in S_i$ has $I(b) = \{i\}$. Therefore $\beta_i = 2/q$ for all i and

$$1 = \sum_{i=1}^m \beta_i = \frac{2m}{q},$$

i.e. $q = 2m$, contradicting that q is odd. Hence some $c \in C$ has $|R(c)| \geq \frac{q-1}{q}|A|$, and by (2) $\psi(x, y) \geq 1 - \frac{1}{q}$.

Upper bound. Let $A = \{a_i : i \in \mathbb{Z}_q\}$, $B = \{b_i : i \in \mathbb{Z}_q\}$, $C = \{c_i : i \in \mathbb{Z}_q\}$, put $S_i = \{b_i, b_{i+1}\}$ (indices modulo q ; $b_i \neq b_{i+1}$ as $q \geq 3$), and let the graph have exactly the edges given by

$$N_B(a_i) = B \setminus S_i, \quad N_B(c_i) = S_i \quad (i \in \mathbb{Z}_q).$$

This is a tripartition. Each a_i has $q - 2 = x|B|$ neighbours in B ; b_j lies in S_j and S_{j-1} only, so it is non-adjacent to exactly a_j, a_{j-1} and has $q - 2 = x|A|$ neighbours in A ; each c_i has $2 = y|B|$ neighbours in B ; and b_j has exactly the two neighbours c_j, c_{j-1} in C , i.e. $2 = y|C|$. So the triple is (x, y) -biconstrained. The vertex c_i does not reach a_i , since $N_B(a_i) \cap S_i = \emptyset$. For $j \neq i$ the two-element sets $S_i \neq S_j$ satisfy $S_i \setminus S_j \neq \emptyset$, so $N_B(c_i) \cap N_B(a_j) = S_i \setminus S_j \neq \emptyset$ and c_i reaches a_j . Hence every c_i reaches exactly $q - 1$ vertices of A , and $\psi(x, y) \leq 1 - \frac{1}{q}$. $\square$

Proposition 3.1. *For every odd $q \geq 7$, (A, B, C) is a $(2/q, (q-2)/q)$ -biconstrained tripartition of G_q with $|A| = q(q+1)/2$, $|B| = |C| = q$, and every $c \in C$ satisfies $|A \setminus R(c)| = q - 2$. Consequently*

$$\psi\left(\frac{2}{q}, \frac{q-2}{q}\right) \leq 1 - \frac{q-2}{q(q+1)/2} = 1 - \frac{2(q-2)}{q(q+1)}.$$

Proof. Throughout, $x = 2/q$ and $y = (q-2)/q$. First, $|B| = 2h + 3 = q$ and $|C| = 2h + 3 = q$ by construction, and

$$|A| = m(q-2) + 3 = \frac{(q+3)(q-2)}{2} + 3 = \frac{q^2 + q - 6}{2} + 3 = \frac{q(q+1)}{2}.$$

The sets $S_1, \dots, S_m$ are pairwise distinct two-element subsets of B : the pairs P_i are disjoint from each other and from $\{r_{12}, r_{13}, r_{23}\}$, and the three sets $S_{h+1}, S_{h+2}, S_{h+3}$ are the three two-subsets of $\{r_{12}, r_{13}, r_{23}\}$.

A-B conditions. A vertex $a \in A_i$ has $|N_B(a)| = |S_i| = 2 = \frac{2}{q}|B| = x|B|$; a vertex $u \in U$ has $|N_B(u)| = q \geq x|B|$. A vertex $b \in P_i$ belongs to S_i and to no other S_j , so $N_A(b) = A_i \cup U$ and $|N_A(b)| = (q-2) + 3 = q+1 = \frac{2}{q} \cdot \frac{q(q+1)}{2} = x|A|$. A vertex r_{jk} belongs to exactly two of $S_{h+1}, S_{h+2}, S_{h+3}$ (namely those containing r_{jk}) and to no P_i , so $|N_A(r_{jk})| = 2(q-2) + 3 = 2q-1 \geq q+1 = x|A|$.

B-C conditions. A vertex c of type i has $|N_B(c)| = |B \setminus S_i| = q-2 = \frac{q-2}{q}|B| = y|B|$. A vertex $b \in B$ is non-adjacent to c exactly when $b \in S_{\text{type}(c)}$. If $b \in P_i$, this happens exactly for the two vertices of type i ; if $b = r_{jk}$, it happens for the two types among $h+1, h+2, h+3$ whose set contains r_{jk} , each of which occurs once. So every b is non-adjacent to exactly two vertices of C and $|N_C(b)| = q-2 = y|C|$. Thus (A, B, C) is (x, y) -biconstrained.

Second neighbourhoods. Let c have type i , so $N_B(c) = B \setminus S_i$. A vertex $a \in A_i$ has $N_B(a) = S_i$, disjoint from $N_B(c)$, so $a \notin R(c)$. For $j \neq i$, $S_j \neq S_i$ are two-element sets, so $S_j \setminus S_i \neq \emptyset$, i.e. every $a \in A_j$ has a common neighbour with c in $B \setminus S_i$; and every $u \in U$ is adjacent to all of $B \setminus S_i \neq \emptyset$. Hence $A \setminus R(c) = A_i$ and $|A \setminus R(c)| = q-2$, for every $c \in C$. Every c therefore reaches the fraction

$$\frac{|A| - (q-2)}{|A|} = 1 - \frac{q-2}{q(q+1)/2} = 1 - \frac{2(q-2)}{q(q+1)}$$

of A , and (2) gives the stated upper bound on $\psi(x, y)$. $\square$

Theorem 1.2 (Main theorem). *For every odd integer $q \geq 7$,*

$$\psi\left(\frac{2}{q}, \frac{q-2}{q}\right) \leq 1 - \frac{2(q-2)}{q(q+1)} < 1 - \frac{1}{q} = \psi\left(\frac{q-2}{q}, \frac{2}{q}\right).$$

Proof. Let $q \geq 7$ be odd. Proposition 3.1 gives $\psi(2/q, (q-2)/q) \leq 1 - \frac{2(q-2)}{q(q+1)}$ and Lemma 2.1 gives $\psi((q-2)/q, 2/q) = 1 - \frac{1}{q}$. By (3), $\frac{2(q-2)}{q(q+1)} > \frac{1}{q}$ for $q > 5$, so $1 - \frac{2(q-2)}{q(q+1)} < 1 - \frac{1}{q}$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/1902.10878__01/`. Treat it as machine output, not as an authority.

Claim under review: the writeup at `attacks/1902.10878__01/output.md` claims to **disprove** the open question “does $\psi(x,y) = \psi(y,x)$ for all x,y in $(0,1]$?” from arXiv:1902.10878 (Chudnovsky, Hompe, Scott, Seymour, Spirk, *Concatenating bipartite graphs*), via

- a Lemma: $\psi((q-2)/q, 2/q) = 1 - 1/q$ for every odd $q \geq 3$, and
- a construction: $\psi(2/q, (q-2)/q) \leq 1 - 2(q-2)/(q(q+1))$ for odd $q \geq 7$,

which for $q = 7$ gives $\psi(2/7, 5/7) \leq 23/28 < 24/28 = 6/7 = \psi(5/7, 2/7)$.

B.1 Interpretation

I downloaded the arXiv v3 TeX source (Dec 2020, the version into which the withdrawn companion arXiv:1908.07453 was merged) and checked the definitions verbatim.

- Paper definition (source lines 101–110, 179–196): a *tripartition* (A,B,C) is a partition of $V(G)$ into nonempty **stable** sets; (x,y) -constrained means every a in A has $\geq x|B|$ neighbours in B , every b in B has $\geq y|C|$ neighbours in C , and **no edges between A and C**; (x,y) -*biconstrained* adds: every b in B has $\geq x|A|$ neighbours in A and every c in C has $\geq y|B|$ neighbours in B . $\psi(x,y)$ is the maximum z such that every (x,y) -biconstrained G has some v in C with $|N^2_A(v)| \geq z|A|$, where N^2 is distance **exactly** two.
- The writeup’s four degree conditions match the paper’s exactly (same sides, same fractions, no silent strengthening or weakening). Its $R(c) = \{a : N_B(a) \text{ and } N_B(c) \text{ intersect}\}$ equals $N^2_A(c)$: in a tripartition $N(a), N(c)$ are subsets of B and there are no A – C edges, so distance exactly 2 from c to a is equivalent to a common B -neighbour. The writeup’s constructions have all three parts stable, no A – C edges, and $V = A + B + C$, so they are legitimate tripartitions.
- The open question is stated in the paper as a prose remark right after Theorem 2.3 (permute, $\psi(x,y)=\psi(y,x)$), source line 437: “we have not been able to prove an analogue of [2.3] for the biconstrained case, or for the exact case, although we have no counterexample for either one.” The catalog question (“Does $\psi(x,y)=\psi(y,x)$ hold for all x,y in $(0,1]$?”) is exactly what the writeup answers, in the negative, with a concrete counterexample pair. No interpretation gaming: a reasonable author of the original paper would consider the ψ -symmetry question resolved (negatively) by this argument. The writeup correctly does not claim anything about the exact-case (xi) symmetry, which remains open.

Interpretation: **the intended one**.

B.2 Step-by-step findings

step	label	note
1.1 Definitions of biconstrained and ψ	VALID	Matches arXiv v3 verbatim (lines 101–110, 179–196 of source). inf/max formulation equivalent to the paper's "maximum z " (paper proves the max exists, Thm suptomax).
2.1 Lemma lower bound: disjointness forces equality chain (1)	VALID	If a is not in $R(c)$, $N_B(a)$, $N_B(c)$ disjoint; degree sums give $ B \geq ((q-2)/q + 2/q) B = B $, so all equalities and $N_B(a) = B \setminus N_B(c)$. Under the contradiction hypothesis every c has $ A /q \geq 1$ unreachable a , so every c has exactly $(2/q) B $ B -neighbours. (Divisibility is not an issue: equality forced means $(2/q) B $ is an integer; if it could not be, no c has an unreachable vertex, contradicting the hypothesis directly.)
2.2 Edge count gives (2): every b has exactly $(2/q) C $ C -neighbours	VALID	$e(B, C) = (2/q) B C $ from the C side; each of the $ B $ terms on the B side is $\geq (2/q) C $; equality in the total forces equality termwise.
2.3 Classes C_i , A_i and (3): $\alpha_i > 1/q$	VALID	For c in C_i the unreachable set is exactly $A_i = \{a : N_B(a) = B \setminus S_i\}$ (both inclusions immediate from (1)); the A_i are pairwise disjoint (distinct S_i); hypothesis $ R(c) < (q-1) A /q$ gives $ A_i > A /q$.
2.4 (4): sum of β_i over i in $I(b)$ equals $2/q$	VALID	$\{c : b \in N_B(c)\} = N_C(b)$, which has size $(2/q) C $ by (2). Hence $I(b)$ nonempty.
2.5 (5): sum of α_i over i in $I(b)$ at most $2/q$	VALID	$b \in S_i$ means A_i is entirely nonadjacent to b ($N_B(a) = B \setminus S_i$); b has $\geq (q-2) A /q$ A -neighbours (biconstrained, $x = (q-2)/q$), so at most $2 A /q$ nonneighbours; the A_i are disjoint.

step	label	note
2.6 Parity contradiction: $ I(b) = 1$, $\beta_i = 2/q$, $2m = q$	VALID	Two alpha-terms each $> 1/q$ would exceed $2/q$, so $ I(b) = 1$; then (4) gives $\beta_i = 2/q$; every i occurs (S_i nonempty, pick b in S_i); summing, $1 = 2m/q$, impossible for odd q . Finite core machine-verified (script <code>verify_lemma_core.py</code>): no matching of K_q covers all q vertices for $q = 3..13$ (exhaustive enumeration, e.g. 232 matchings of K_7 , 568504 of K_{13} ; sanity check $q = 6$ does have one). Hence $\psi((q-2)/q, 2/q) \geq 1 - 1/q$. Verified by computer for $q = 3, 5, 7, 9, 11, 13$: the triple is exactly $((q-2)/q, 2/q)$ -biconstrained and every c in C has $ N^2_A(c) = q-1$, so $\psi((q-2)/q, 2/q) \leq 1 - 1/q$. Together with 2.6: equality.
2.7 Lemma upper bound: cyclic construction	VALID	$ B = 2h + 3 = q$; the m 2-subsets are pairwise distinct (disjoint pairs; the three r -sets pairwise distinct and supported off the pairs). Verified in code.
3.1 Set B: h pairs plus r_{12}, r_{13}, r_{23} ; $m = (q+3)/2$ distinct 2-subsets S_i	VALID	$ A = m(q-2) + 3 = q(q+1)/2$ (exact-fraction check passed). A_i -vertices have exactly $2 = (2/q) B $ B-neighbours; pair-vertices have exactly $q+1 = (2/q) A $ A-neighbours (tight); r -vertices have $2q-1 \geq q+1$. Verified by computer for $q = 7, 9, 11$.
3.2 Set A and the A–B side, $ A = q(q+1)/2$	VALID	$ C = 2h + 3 = q$; every c has $q-2 = ((q-2)/q) B $ B-neighbours; every b is omitted by exactly two C-vertices (pair-vertices by the two type- i copies; r -vertices by the two singleton types containing them), so has $q-2 = ((q-2)/q) C $ C-neighbours. Verified by computer. The triple is $(2/q, (q-2)/q)$ -biconstrained.
3.3 Set C and the B–C side	VALID	

step	label	note
3.4 Second neighbourhoods and final inequality	VALID	c of type i misses exactly the $q-2$ vertices of A_i ($S_j \cap S_i$ nonempty for $j \neq i$ since the S 's are distinct 2-sets; the 3 universal vertices are always reached). Computer: for $q = 7/9/11$ every c reaches exactly 23/38/57 of $ A = 28/45/66$. And $2(q-2)/(q(q+1)) - 1/q = (q-5)/(q(q+1)) > 0$ iff $q > 5$ (exact-fraction check).
4 $q = 7$ instance	VALID	Independently rebuilt: $ A = 28$, $ B = C = 7$, biconstrained for $(2/7, 5/7)$, every C-vertex reaches exactly 23 of 28 A-vertices; $23/28 < 24/28 = 6/7$.
Verdict block ("disproved", one_line, caveats)	VALID	Accurate: symmetry fails at $(x,y) = (2/7, 5/7)$; the writeup correctly notes $\text{psi}(2/7, 5/7)$ is not determined exactly and that the exact-degree (xi) question is untouched.

No GAP or ERROR found. The only step that is not a finite object — the Lemma's lower bound (2.1–2.6), a universal statement over all biconstrained graphs — was re-derived line by line above and its combinatorial core (the forced matching structure and the parity obstruction) machine-checked.

B.3 Reference check

The writeup cites no external theorems; everything is self-contained except the definitions, which I verified against the downloaded arXiv v3 TeX source of 1902.10878 (identical to the published EJC version, v29i2p47, 2022, per the journal listing). Additional due diligence — consistency of the two claimed values with everything the source paper proves:

- **trivialbounds** (paper Thm 2.2): $\max(x,y) \leq \text{psi}(x,y) \leq (\text{ceil}(kx) + \text{ceil}(ky) - 1)/k$. For $(5/7, 2/7)$, $k = 7$: psi in $[5/7, 6/7]$; the Lemma's value $6/7$ attains the upper (Cayley) bound — consistent. For $(2/7, 5/7)$: psi in $[5/7, 6/7]$, and $23/28$ lies strictly inside — consistent.
- $3/4\text{ub2}$ ($x > 2/3$, $x+2y > 1$, $4(1-x)(1-y) \leq 1$): gives $\text{psi}(5/7, 2/7) \geq 3/4$ ($40/49 \leq 1$); consistent with (and weaker than) the Lemma's $6/7$.
- $3/4\text{ub4}$ ($x \geq 1/4$, $y > 2/3$): gives $\text{psi}(2/7, 5/7) \geq 3/4 = 21/28$, consistent with the new upper bound $23/28$ (so $\text{psi}(2/7, 5/7)$ is pinned to $[3/4, 23/28]$).
- $3/4\text{psi1b}$ (the paper's $\text{psi} < 3/4$ regions): none of its four bullets covers $(2/7, 5/7)$ or $(5/7, 2/7)$ — no conflict.
- $2+3/3+2$ ($\text{psi}(x, 1/3) \geq 2/3$ for $x > 1/2$, $\text{psi}(1/3, y) \geq 2/3$ for $y > 1/2$): with the trivial bound these give $\text{psi}(1/3, 2/3) = \text{psi}(2/3, 1/3) = 2/3$, matching the Lemma's $q = 3$ case (where the writeup's two bounds coincide and no asymmetry arises — the writeup's condition $q \geq 7$ is exactly right, the gap $(q-5)/(q(q+1))$ vanishing at $q = 5$).
- **peacepsi** (paper Thm 10.2, the paper's own partial symmetry: $\text{psi}(x,y) = x$ iff $\text{psi}(y,x) = x$): since $\text{psi}(5/7, 2/7) = 6/7 \neq 5/7$, it forces $\text{psi}(2/7, 5/7) > 5/7$ — consistent with $[3/4, 23/28]$, and confirms the counterexample lives precisely where **peacepsi** has no grip (neither value equals $\max(x,y)$).

- Literature search (Sept 2026): no follow-up resolving psi-symmetry was found; the EJC 2022 version still lists the question as open, matching the catalog’s review. The Lemma’s exact value $\psi((q-2)/q, 2/q) = 1 - 1/q$ does not appear in the paper (its best applicable bound there is $3/4$), so the result is new, not ALREADY_KNOWN.

Sources: [arXiv:1902.10878](https://arxiv.org/abs/1902.10878), [EJC v29i2p47](#), [Princeton copy](#).

B.4 Computational check

Scripts in `verification/scripts/1902.10878__01/` (plain python3, exact fractions arithmetic, no external packages):

- `verify_constructions.py` — rebuilds both constructions independently from the writeup’s prose and checks every degree condition against the paper’s definitions (stability and no A–C edges hold by construction and are respected in the distance-2 computation). Output:
 - Section-3 construction, $q = 7$: $|A| = 28$, $|B| = |C| = 7$, biconstrained = YES for $(2/7, 5/7)$, $|N^2_A(c)| = 23$ for **every** c in C , ratio exactly $23/28$. $q = 9$: $|A| = 45$, all c reach 38 ($38/45$). $q = 11$: $|A| = 66$, all c reach 57 ($57/66$). All match the claimed $1 - 2(q-2)/(q(q+1))$.
 - Cyclic construction, $q = 3, 5, 7, 9, 11, 13$: biconstrained = YES for $((q-2)/q, 2/q)$, every c reaches exactly $q-1$ of q , ratio $1 - 1/q$. For $q = 7$: $6/7 = 24/28$.
 - ALL CHECKS PASSED.
- `verify_lemma_core.py` — machine check of the finite core of the Lemma’s lower bound: (a) exact-fraction verification that two terms $> 1/q$ exceed $2/q$ (forcing $|I(b)| = 1$); (b) exhaustive enumeration of all matchings of K_q for $q = 3..13$ confirming none covers all q vertices (4, 26, 232, 2620, 35696, 568504 matchings respectively; sanity check at $q = 6$ correctly finds a perfect matching); (c) exact re-derivation of every displayed identity ($|A| = q(q+1)/2$, $q+1 = (2/q)|A|$, $\text{gap} = (q-5)/(q(q+1))$, $1 - 10/56 = 23/28 < 24/28 = 6/7$). ALL CHECKS PASSED.

No claimed property failed; every number in the writeup reproduced exactly.

B.5 Caveats

- The Lemma’s lower bound $\psi(5/7, 2/7) \geq 6/7$ is a statement over all biconstrained graphs and cannot be brute-forced; it rests on the hand-checked rigidity argument (steps 2.1–2.6). I re-derived each forced equality myself and found no gap; the final obstruction is machine-verified. This is the one load-bearing part not reducible to a finite certificate.
- The writeup determines $\psi(5/7, 2/7) = 6/7$ exactly but only bounds $\psi(2/7, 5/7)$ from above ($23/28$); combined with the paper’s $3/4 \leq$ the true value lies in $[21/28, 23/28]$. The disproof needs only $23/28 < 24/28$, which is airtight; the writeup’s caveat says this correctly.
- The asymmetry genuinely requires $q \geq 7$: at $q = 3$ and $q = 5$ the two bounds coincide ($\text{gap} = (q-5)/(q(q+1)) = 0$ or negative), consistent with the paper’s known equalities at $(1/3, 2/3)$. No off-by-one issues found.
- Convention dependence checked: “distance exactly two” vs “common B-neighbour” coincide here because tripartitions forbid A–C edges and internal edges; the writeup’s constructions satisfy the tripartition requirements (nonempty stable parts, partition of V).
- Divisibility edge cases in the Lemma (whether $(2/q)|B|$, $(2/q)|C|$, $|A|/q$ are integers) are all handled by the forced-equality structure, as noted in step 2.1.
- Untouched by the writeup, and correctly flagged as such: phi-symmetry (paper’s Thm 2.3) is unaffected, and the exact-case (xi) symmetry question remains open. The construction’s three A-vertices complete to B mean it says nothing about the exact-degree regime.

B.6 Referee summary

I set out to break this proof and could not. The claimed counterexample splits into a finite certificate and one infinite-family lemma. The finite certificate — a 42-vertex $(2/7, 5/7)$ -biconstrained triple with $|A| = 28$, $|B| = |C| = 7$ in which every C-vertex reaches exactly 23 of the 28 A-vertices — was rebuilt independently from the prose and verified exactly (as were the $q = 9, 11$ members of the family and the cyclic constructions), against the definitions taken verbatim from the arXiv v3 source of 1902.10878. The lemma $(\psi(5/7, 2/7) = 6/7)$ I verified line by line: the disjointness/equality rigidity, the edge-count, and the packing inequalities are all correct, and the concluding parity obstruction was additionally machine-checked by exhaustive matching enumeration. The two values are consistent with every theorem the source paper proves (trivial bounds, the $3/4$ -threshold theorems, and notably peacepsi , the paper's own partial symmetry result, which the counterexample cleanly evades), the question is confirmed open in the literature as of September 2026, and the interpretation is exactly the authors' question. The symmetry $\psi(x, y) = \psi(y, x)$ is disproved.

References

- [1] M. Chudnovsky, P. Hompe, A. Scott, P. Seymour and S. Spirkl, Concatenating bipartite graphs, *Electron. J. Combin.* 29(2) (2022), P2.47; arXiv:1902.10878.